# Chapter 1

# Homological algebra

# Chapter 2

# Abelian categories

## 2.1 Additive categories

**Definition 2.1.1** (Additive Category)**.** A Category $\mathsf{A}$ is **additive** if it verifies:

A.1) There exists a Zero object $0 \in \mathsf{A}$.

A.2) $\mathsf{A}$ has finite Products.

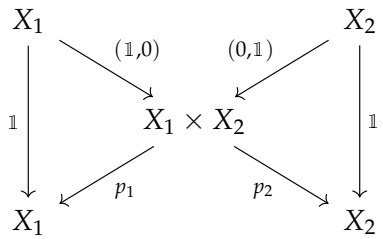
A.3) For every pair of objects $X, Y \in \mathsf{A}$ the set $\mathrm{Hom}_{\mathsf{A}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.

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**Lemma 2.1.1.** *Finite Products and Coproducts coincide in an Additive Category.*

*Proof.* Consider $X_1, X_2 \in \mathsf{A}$. We only need to check that $X_1 \times X_2 \cong X_1 \amalg X_2$ since the case for a finite collection of elements follows directly by induction. To do this, we will see that $X_1 \times X_2$ satisfies the universal property of the Coproduct.

Consider the following diagram



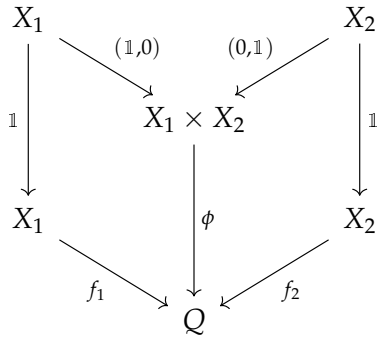
Note that for any pair of morphisms $(\alpha_1, \beta_1), (\alpha_2, \beta_2) : R \to X_1 \times X_2$ we have that

$$(\alpha_1, \beta_1) + (\alpha_2, \beta_2) = (\alpha_1 + \alpha_2, \beta_1, \beta_2)$$

Therefore if $p_1, p_2$ are two projections on $X_1 \times X_2$ whose sum is the identity we have that

$$(\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2 = \mathbb{1}_{X_1 \times X_2}$$

Consider now two morphisms $f_i : X_i \to Q$. Then its sum $\phi = f_1 \circ p_1 + f_2 \circ p_2$ produces the following diagram




Furthermore, this is unique. Indeed, consider $\phi' : X_1 \times X_2 \to Q$ another morphism filling the diagram. Then

$$\begin{aligned}
\phi &= \phi \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi \circ (id, 0) \circ p_1 + \phi \circ (0, \mathbb{1}) \circ p_2 \\
&= f_1 \circ p_1 + f_2 \circ p_2 \\
&= \phi' \circ (\mathbb{1}, 0) + \phi' \circ (0, \mathbb{1}) \\
&= \phi' \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi' \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi'
\end{aligned}$$

Therefore $X_1 \times X_2$ verifies the universal property of $X_1 \amalg X_2$. □

*Notation.* If $\{X_i\}_{i=1}^n$ is a finite set of objects in an Additive Category we will denote its Product (and therefore by Lemma 2.1.1 its Coproduct) by $\bigoplus_{i=1}^n X_i$.

**Lemma 2.1.2.** *A morphism $\varphi : A \to B$ in an Additive Category is a monomorphism if and only if for every morphism $\xi : Z \to A$*

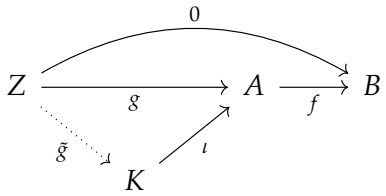
$$\varphi \circ \xi = 0 \implies \xi = 0$$

*Dually $\varphi$ is an epimorphism if and only if for every morphism $\xi : B \to Z$*

$$\xi \circ \varphi = 0 \implies \xi = 0$$

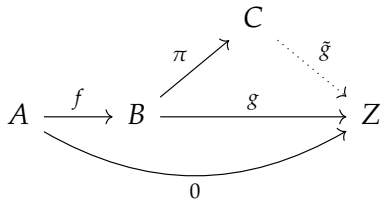
**Definition 2.1.2** (Additive Functor)**.** Let $\mathsf{A}, \mathsf{B}$ be two Additive Categorys. A Functor $F : \mathsf{A} \to \mathsf{B}$ is **additive** if it preserves the abelian group structure of all Hom-sets. ⌟

**Definition 2.1.3** (Kernel)**.** Let $f : A \to B$ be a morphism in an Additive Category $\mathsf{A}$. A morphism $\iota : K \to A$ is a **kernel** of $f$ if $f \circ \iota = 0$ and for every morphism $g : Z \to A$ such that $f \circ g = 0$ there exists a unique $\tilde{g} : Z \to K$ making the following diagram commute




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**Definition 2.1.4** (Cokernel)**.** Let $f : A \to B$ be a morphism in an Additive Category $\mathsf{A}$. A morphism $\pi : B \to C$ is a **cokernel** of $f$ if $\pi \circ f = 0$ and for every morphism $g : B \to Z$ such that $\pi \circ g = 0$ there exists a unique $\tilde{g} : C \to Z$ making the following diagram commute

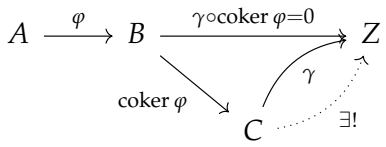



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**Lemma 2.1.3.** *In an Additive Category Kernels are monomorphisms and Cokernels are epimorphisms.*

*Proof.* We will only show that Cokernels are epimorphisms. The statement about Kernels follows analogously.

Let $\varphi : A \to B$ be a morphism and let $\operatorname{coker} \varphi : B \to C$ be its Cokernel. Consider another morphism $\gamma : C \to Z$ such that $\gamma \circ \operatorname{coker} \varphi = 0$. Since $(\gamma \circ \operatorname{coker} \varphi) \circ \varphi = 0$ by definition of the Cokernel we have




Therefore since $\gamma \circ \operatorname{coker} \varphi = 0 = 0 \circ \operatorname{coker} \varphi$ uniqueness forces $\gamma = 0$ and thus by Lemma 2.1.2 $\varphi$ is an epimorphism. □

**Lemma 2.1.4.** *Let $\varphi : A \to B$ be a morphism in an Additive Category. Then $\varphi$ is a monomorphism if and only if $0 \to A$ is its Kernel and $\varphi$ is an epimorphism if and only if $B \to 0$ is its Cokernel.*

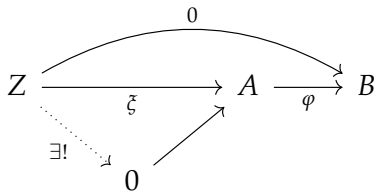
*Proof.* We will do the Kernel one.

$\Longrightarrow$ Suppose $\varphi$ is a monomorphism.

If $\xi : Z \to A$ is a morphism such that $\varphi \circ \xi = 0$ then by Lemma 2.1.2 $\xi = 0$. In particular $\xi$ factors uniquely through $0 \to A$, i.e: $0 \to A$ is the Kernel of $\varphi$.

$\Longleftarrow$ Suppose now that $0 \to A$ is the Kernel of $\varphi$.

Consider a morphism $\xi : Z \to A$ such that $\varphi \circ \xi = 0$. By the Kernel properties we have the following diagram




Therefore by uniqueness $\xi = 0$, making $\varphi$ a monomorphism by Lemma 2.1.2.

□

## 2.2 Abelian categories

**Definition 2.2.1** (Abelian Category)**.** An Additive Category A is **abelian** if it verifies

A.4) Kernels and Cokernels exist in A.

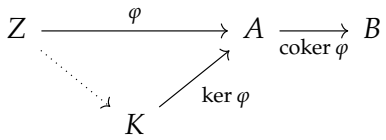
Furthermore every monomorphism is the Kernel of some morphism and every epimorphism is the Cokernel of some morphism.

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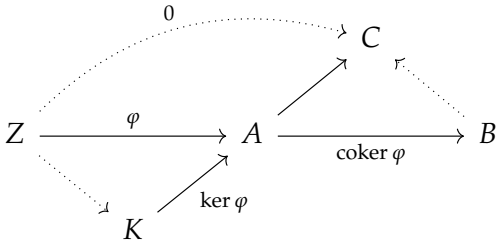
**Lemma 2.2.1.** *In an Abelian Category A every Kernel is the Kernel of its Cokernel and every Cokernel is the Cokernel of its Kernel.*

*Proof.* Lets check Cokernels this time.

Let coker $\varphi : A \to B$ be the Cokernel of a morphism $\varphi : Z \to A$. Since A is abelian $\varphi$ has a kernel $\ker \varphi : K \to A$. Note that $Z \to A \to B$ is 0, therefore by the Kernel universal property we have




Consider a morphism $A \to C$ such that $(K \to A \to C) = 0$. Now using the Cokernel universal property we obtain

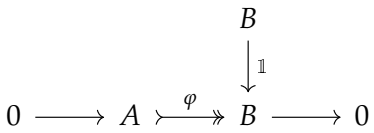



Therefore coker $\varphi$ satisfies the universal property of the Cokernel of $\ker \varphi : K \to A$. □

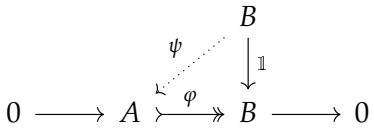
**Lemma 2.2.2.** *Let $\varphi : A \to B$ be a morphism in an Abelian Category* A. *If $\varphi$ is both a monomorphism and epimorphism then $\varphi$ is an Isomorphism.*

*Proof.* By Lemma 2.1.4 the Kernel of $\varphi$ is $0 \to A$ and the Cokernel is $B \to 0$. Furthermore by Lemma 2.2.1 $\varphi$ is the Cokernel of $0 \to A$ and the Kernel of $B \to 0$.

Consider the following diagram

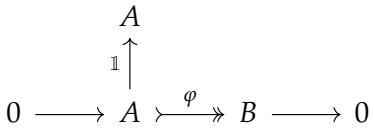



Since $(B \to B \to 0) = 0$ and $\varphi$ is the Kernel of $B \to 0$ we have

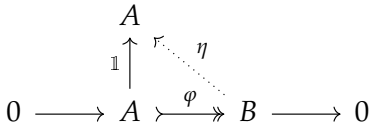



Therefore since $\varphi\psi = \mathbb{1}$ we have shown that $\varphi$ has a right-inverse.

Consider now the diagram



Since $(0 \to A \to A) = 0$ and $\varphi$ is the Cokernel of $0 \to A$ we have



Therefore since $\eta\varphi = \mathbb{1}$ we have shown that $\varphi$ has a left-inverse. □

**Theorem 2.2.1** (Freyd-Mitchell theorem)**.** *Let* A *be a small Abelian Category. Then there exists a fully faithfull* `def:exact-functor` $\mathsf{A} \to R\text{-}\mathsf{Mod}$ *for a suitable (possibly non-commutative) ring* $R$*, where* $R$-Mod *is the category of left* $R$*-modules.*

## 2.3 Complexes and (co)homology

From now on, we will be working in a fixed Abelian Category A. By the Freyd-Mitchell theorem we can work with objects and morphisms using set-theoretic tools when needed.

**Definition 2.3.1** (Complex)**.** A sequence of objects and morphisms in an Additive Category A

$$\cdots \xrightarrow{d_{i-2}} C_{i-1} \xrightarrow{d_{i-1}} C_i \xrightarrow{d_i} C_{i+1} \xrightarrow{d_{i+1}} \cdots$$

is a **chain complex** if for every $i \in \mathbb{N} \quad d_i \circ d_{i-1} = 0$. We denote a chain complex as $(C_\bullet, d_\bullet)$ or simply as $C_\bullet$.

By reversing the order, we say that a sequence

$$\cdots \xrightarrow{d^{i+2}} C^{i+1} \xrightarrow{d^{i+1}} C^i \xrightarrow{d^i} C^{i-1} \xrightarrow{d^{i-1}} \cdots$$

is a **cochain complex** if for every $i \in \mathbb{N} \quad d_i \circ d_{i+1} = 0$. We denote a cochain complex as $(C^\bullet, d^\bullet)$ or simply as $C^\bullet$. ⌟

*Remark* 2.3.1. In purelly algebraic terms, chains are cochains are *equivalent*. We will adopt the cochain notation. ▽

**Definition 2.3.2** (Exact complex)**.** A Complex in an Abelian Category

$$\cdots \xrightarrow{d^{i+2}} C^{i+1} \xrightarrow{d^{i+1}} C^i \xrightarrow{d^i} C^{i-1} \xrightarrow{d^{i-1}} \cdots$$

is **exact at** $C^i$ (or simply at position $i$) if

$$\operatorname{coker} d^{i+1} \circ \ker d^i = 0$$

A complex is **exact** if it is exact at every position. ⌟

**Definition 2.3.3** (Image and coimage)**.** Let $\varphi : A \to B$ be a morphism in an Abelian Category.

The **image** of $\varphi$ is $\operatorname{im} \varphi = \ker(\operatorname{coker} \varphi)$ and the **coimage** of $\varphi$ is $\operatorname{coim} \varphi = \ker(\operatorname{coker} \varphi)$. ⌟

**Definition 2.3.4.** Let $C^\bullet$ be a Complex in an Abelian Category $\mathsf{A}$. We define the $i$**-th cohomology group** of $C^\bullet$ as

$$\mathrm{H}^i(C^\bullet) = {\ker d^i}/{\operatorname{im} d^{i-1}}$$

⌟

**Definition 2.3.5** (Category of complexes)**.** Let $\mathsf{A}$ be an Abelian Category. The **Category of Complexes** $\mathsf{C}(\mathsf{A})$ consists of:

1. Cochain complexes in $\mathsf{A}$ as objects.
2. For $(M^\bullet, d^\bullet_{M^\bullet}), (N^\bullet, d^\bullet_{N^\bullet}) \in \mathsf{C}(\mathsf{A})$, $\operatorname{Hom}_{\mathsf{C}(\mathsf{A})}(M^\bullet, N^\bullet)$ consists of the commutative diagrams of the form

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & M^{i-1} & \xrightarrow{d^{i-1}_{M^\bullet}} & M^i & \xrightarrow{d^i_{M^\bullet}} & M^{i+1} & \xrightarrow{d^{i+1}_{M^\bullet}} & \cdots \\
 & & \downarrow{\scriptstyle \alpha^{i-1}} & & \downarrow{\scriptstyle \alpha^i} & & \downarrow{\scriptstyle \alpha^{i+1}} & & \\
\cdots & \longrightarrow & N^{i-1} & \xrightarrow{d^{i-1}_{N^\bullet}} & N^i & \xrightarrow{d^i_{N^\bullet}} & N^{i+1} & \xrightarrow{d^{i+1}_{N^\bullet}} & \cdots
\end{array}$$

   We denote by $\alpha^\bullet : M^\bullet \to N^\bullet$ the morphism determined by the collection of $\alpha^i$.

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**Lemma 2.3.1.** *The Category of complexes of an Abelian Category is an Abelian Category.*

**Lemma 2.3.2.** *Let $\mathsf{A}$ be a Abelian Category and let $\mathsf{C}(\mathsf{A})$ be its Category of complexes.*

*Then for every integer $i$, the assignement $\mathrm{H}^i : M^\bullet \mapsto H^i(M^\bullet)$ taking a Complex $M^\bullet$ to its $i$-th is a Functor $\mathsf{C}(\mathsf{A}) \to \mathsf{A}$.*

*Proof.* We want to show that for all Complexes $M^\bullet, N^\bullet \in \mathsf{C}(\mathsf{A})$ and $i \in \mathbb{Z}$, $\mathrm{H}^i$ induces in a natural way a homomorphism of abelian groups

$$\operatorname{Hom}_{\mathsf{C}(\mathsf{A})}(M^\bullet, N^\bullet) \to \operatorname{Hom}_{\mathsf{A}}(\mathrm{H}^i(M^\bullet), \mathrm{H}^i(N^\bullet))$$

Let $\alpha^\bullet : M^\bullet \to N^\bullet$ be a morphism in $\mathsf{C}(\mathsf{A})$. Let's consider the relevant of the commutative diagram defining $\alpha^\bullet$:

$$\begin{array}{ccccc} M^{i-1} & \xrightarrow{d^{i-1}_{M^\bullet}} & M^i & \xrightarrow{d^i_{M^\bullet}} & M^{i+1} \\ \downarrow{\scriptstyle \alpha^{i-1}} & & \downarrow{\scriptstyle \alpha^i} & & \downarrow{\scriptstyle \alpha^{i+1}} \\ N^{i-1} & \xrightarrow{d^{i-1}_{N^\bullet}} & N^i & \xrightarrow{d^i_{N^\bullet}} & N^{i+1} \end{array}$$

By the commutativity of the right square we have that $d^i_{N^\bullet} \circ \alpha^i = \alpha^{i+1} \circ d^i_{M^\bullet}$ is 0 on $\ker d^i$. Therefore the restriction of $\alpha^i$ to $\ker d^i_{M^\bullet}$ factors through $\ker d^i_{N^\bullet}$. Composing with the projection we get a morphism

$$\ker d^i_{M^\bullet} \to \mathrm{H}^i(N^\bullet)$$

Since this morphism is 0 when restricted to $\operatorname{im} d^{i-1}_{M^\bullet}$ by the commutativity of the left square we get that $\alpha^i$ induces a morphism

$$\mathrm{H}^i(M^\bullet) \to \mathrm{H}^i(N^\bullet)$$

in A as we wanted. □

*Remark* 2.3.2. We can view  as a Functor $\mathsf{C}(\mathsf{A}) \to \mathsf{C}(\mathsf{A})$ by placing each  object $\mathrm{H}^i(M^\bullet)$ in degree $i$ and connecting them using 0 morphisms, obtaining a complex $\mathrm{H}^\bullet(M^\bullet)$.

Note that the  of this particular Complex verifies

$$\mathrm{H}^\bullet(\mathrm{H}^\bullet(M^\bullet)) = \mathrm{H}^\bullet(M^\bullet)$$

▽